

Methodology Review — faults & naivities in the experimental program

A code-level audit of how the experiments are set up and what the headline findings actually rest on. Ordered by how load-bearing the fault is. Each item: **what** — why it biases the conclusion — **fix**. Source files cited inline. PDF version:

[diary/methodology-review.pdf](#) (generator: [docs/methodology_review_pdf.py](#)).

Scope note: many of these do **not** mean a finding is wrong — they mean the finding is not *yet* isolated from a confound, or is reported more strongly than the protocol earns.

The robust results (transfer-fails F7, the qualitative width>depth trend) are flagged as such at the end.

CRITICAL — these undercut headline findings

C1 — One base checkpoint per architecture; the shape→prunability program has no base-net variance

[scripts/hypernetwork/shape_studies/iso_accuracy_retrain.py](#) loads a single fixed `.pt` per shape (wide / medium / deep / narrow). The "2 seeds" reseed only the **pruner** (`set_seed(seed)` at line 167, then a fresh `BiLSTMPruner`). The base net is never retrained.

- **Consequence.** F4 / F10 / F11 ("wide 84.6% > medium 76.1% > deep 60.2%") compare

one wide training run against *one* deep run. The width/depth gap could be the particular init + SGD trajectory of those four checkpoints, not the architecture.

- The reported $\pm$ is the half-range over pruner inits of a **near-deterministic** readout,

so it structurally cannot see the dominant variance source (base-net training).

- **Fix.** ≥ 3 independently trained base nets per shape; report mean $\pm$ std over base

seeds, with the pruner retrained on each. This is the single most important change.

C2 — No held-out validation anywhere: the test set selects the operating point AND reports it

`best_within_budget()` picks, per seed, the most-aggressive `sw` whose **test** drop $\leq 2pp$, then reports that same test number as "survivors@2pp":

```
survivors@2pp = min_sw { survivors(sw) : drop_test(sw)  $\leq$  2pp }
```

- **Consequence.** An argmin over a noisy constraint *checked on the reported set* is optimistically biased — you keep whichever `sw`'s test-noise dipped under 2pp. The same

pattern recurs in `dim_sweep.py` and the λ sweeps: the test set is the only held-out data.

- **Fix.** MNIST has 60k train images — carve a 5–10k validation split, select `sw /  $\lambda$` on

it, report drop on the untouched test set.

C3 — The RL agent is trained and selected on the test set

In `src/rl/env.py`, `eval_x`, `eval_y` come from `test_loader`

(`multi_seed_compare.build_fixed_batches(test_loader, ...)`). `orig_acc`, the per-step reward, and `final_acc` are all computed on those 256 **test** samples. `best_greedy_gates` is then chosen by `g_info["final_acc"]` (greedy episode selected by test accuracy) and reported on the full test set.

- **Consequence.** The BiLSTM-vs-RL comparison is not apples-to-apples: BiLSTM's signal is train minibatches, RL's signal *is* the test set. RL has strictly more information and still loses — so the **direction** of F5 is conservative/safe, but it must not be described as a fair head-to-head. Reward + greedy selection on only 256 points → noisy objective and noisy selection, reported on 10k.
- **Fix.** Calibrate/reward on a train-derived split; reserve test purely for the final number.

C4 — $\gamma = 1.0$ is hardcoded, and the F6 "MDP is degenerate" theorem rests on exactly that

`RL_GAMMA = 1.0` in every RL script. With the telescoping reward $r_t = \text{acc}_t - \text{acc}_{t+1}$ the return is $\sum r_t = \text{acc}_{\text{final}} - \text{acc}_{\text{orig}}$, which is **path-independent only because $\gamma = 1$** . With $\gamma < 1$ the discounted return $\sum \gamma^t r_t$ no longer telescopes — ordering matters and the MDP is non-degenerate.

- **Consequence.** F6 is really "REINFORCE with $\gamma = 1$ and a telescoping reward has no ordering gradient" — a statement about a degenerate reward/discount *pair*, not about RL pruning in general. The diary already flags γ as the load-bearing default (CLAUDE.md), but the scripts producing F5/F6 never run $\gamma < 1$, and "RL track is CLOSED" overstates what was tested.
- **Fix.** A γ -sweep on the **frozen** net (not just prune-during-training) is the missing control that would actually close, or reopen, the RL line.

SIGNIFICANT — wrong or misleading metrics

S1 — `train_pruner.py` reports trajectory-averaged train-minibatch numbers as "RESULTS"

`write_summary` computes `avg_drop`, `avg_orig_acc`, `avg_pruned_acc` as means over ****all** 1000 steps **on 64-sample train**** batches (lines 200–202), averaging in the early steps

where nothing was pruned yet, on training data, under a "RESULTS / Avg acc drop" heading.

Only `avg_gate[-1]` is a converged quantity.

- **Fix.** Headline = final mask, evaluated on the test set. (Sweep scripts already do this;

the main documented entry point does not.)

S2 — dim_sweep's "larger nets are more prunable" (F1's 51%) has no accuracy-drop control

`dim_sweep.py` prunes every dim at fixed `SPARSITY_W = 0.05` and reports `pct_pruned` next to **pre-pruning** `te_acc`; post-prune accuracy is never measured.

- **Consequence.** "dim=2048 → 51% pruned" is "% pruned at a fixed penalty," not "%

prunable at iso-accuracy." At fixed absolute sparsity weight a bigger net has more free capacity to surrender at the *same* penalty regardless of redundancy structure, so the trend partly measures the λ -vs-capacity interaction. Single seed, single base-train, no error bars, no drop axis.

- **Fix.** Re-run at iso-accuracy (the F4 protocol). The qualitative trend may survive, but

as run the numbers don't isolate prunability.

S3 — The efficiency metric is dominated by a clamp floor, and cross-net comparisons ride on it

`efficiency_compare.py`: `eff = (% pruned) / max(drop_pp, 0.5)`. Whenever `drop < 0.5pp` the denominator pins at 0.5, so `eff ≈ 2·(% pruned)` — it stops measuring the trade-off. Most MNIST points have `drop < 0.5` (medium $\lambda=0.04 \rightarrow 0.12pp$).

- **Consequence.** MNIST's "peak eff 135.8" vs CIFAR_big's "47.9" (F15) is largely the clamp: MNIST is in the floored regime, CIFAR isn't. Comparing those two numbers as the same quantity, and concluding "task-driven not size-driven," compares a clamped ratio to an unclamped one.

- **Fix.** Report the Pareto curve (% pruned at fixed drop budgets), not a single ratio with a floor.

S4 — The $\lambda_{opt} \cdot N_{layers} \approx 0.097$ "law" is curve-fit on argmax-over-noise, $n \approx 3$

λ_{opt} is the argmax of S3's clamped efficiency over a **coarse** grid

(`{0.04, 0.06, 0.08, ...}`), read to a single grid value per net. When drops sit near the 0.5

floor that argmax is decided by sub-noise % differences. Fitting `$\lambda_{opt} \cdot N_{layers} \approx const$`

across 3 effective datapoints (`$N_L \in \{2,2,3\}$`) and using it to *predict* 1- and 4-layer

λ_{opt} is overfitting a relation with more freedom than constraints.

- **Fix.** Treat as a hypothesis ($n \approx 3$), not a result; the 1-layer and 4-layer datapoints the

diary names as "cheap to test" are exactly the missing constraints.

S5 — "RL loses decisively" (F5) is not supported by the reported stats

BiLSTM 3.68 ± 0.79 vs actor-critic 4.71 ± 1.55 , $n = 5$ each. Welch SE $\approx$

$\sqrt{(0.79^2/5 + 1.55^2/5)} \approx 0.78$, $t \approx 1.03/0.78 \approx 1.3 \rightarrow$ not significant at 5%; the $\pm$ ranges overlap heavily.

- **Consequence.** The broader RL-loses *pattern* across many levers is plausibly real, but "decisively" on this specific comparison isn't earned by $n = 5$. No significance test is run anywhere in the project.
- **Fix.** More seeds and/or a paired test (same base, paired BiLSTM/RL per seed $\rightarrow$ paired t or Wilcoxon).

MODERATE — hygiene / smaller confounds

- **M1 — "2 seeds, BiLSTM near-deterministic" conflates two variances.** Determinism is overpruner init for a *fixed* checkpoint; it says nothing about base-net training variance (C1) or operating-point selection variance (C2). A 2-point half-range is not an error bar.
- **M2 — Cross-net efficiency mixes training budgets.** `efficiency_compare` overlays CIFAR_big (5 epochs) with MNIST (15 epochs) on one axis, and 1-seed "extras" with 3-seed points on one curve. Cross-net $\lambda_{\text{opt/eff}}$ claims compare unequal compute and seed counts.
- **M3 — Hardcoded transcribed numbers.** `efficiency_compare.py` embeds all per- λ results as literals copied from `summary.txt`. These can silently drift from the runs they summarize; read the summaries at runtime instead.
- **M4 — RL state is stale 4 steps of 5.** `_update_activations` runs every `recalibrate_every = 5` steps but the reward is computed every step, so the `mean_act` feature the policy conditions on lags the actual masked network most of the time.
- **M5 — Per-sw reseeding correlates the "independent" runs.** In `iso_accuracy_retrain` `set_seed(seed)` is re-called before every `sw`, so within a seed all `sw` points share identical init and minibatch order — the 5 points are perfectly correlated; only 2 truly independent draws exist.
- **M6 — `train_epoch` runs the model twice per batch with dropout on.** `model(x)` at line 12 (loss) and again at line 16 (accuracy) — the second is a different stochastic pass, so logged train-acc is dropout-perturbed and base training costs $\sim 1.5\times$ more than needed.

Cosmetic (test eval is clean) but it's in every base-train.

What is done well (calibration)

- `evaluate_with_gates` correctly zeros weight **rows and biases** and evaluates on the full 10k test set — final reported drops, *where measured at the converged mask*, are clean.
- Activation interpretability deliberately measures alive/dead firing under the **original** model — the right counterfactual.
- The transfer test (F7) is structurally sound: same layer shapes, frozen-P_A vs oracle-P_B, full-test eval. The transfer-fails conclusion (28pp vs 4pp) is a large effect well outside any of the above biases — **the most robust result in the set**.
- Gradient clipping, tanh-bounded context, +2.0 bias init are reasonable and documented.

The three that matter most

1. **C1** — re-run shape studies with ≥ 3 base-net seeds per architecture before trusting width > depth quantitatively.
2. **C2** — add a validation split; never select sw/λ on the test set you report.
3. **C3 / C4** — the RL "closed" verdict is scoped to $\gamma=1$ + test-set-as-reward; a $\gamma<1$ frozen-net control with a train-derived reward is the honest missing experiment.

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